

FACULTY ELECTRICAL ENGINEERING, MATHEMATICS
AND COMPUTER SCIENCE

Department of Microelectronics

[ET4371] Mixed-mode Wireless transceivers

Assignment 3
Double Balanced Active Mixer

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1 Device Dimensions [Question 1 from the HW]

The goal is to build an active double balanced mixer that achieves an output power of 4dBm and a compression of 35dBc for the third order intermodulation product. As the instruction suggest, we will follow the g_m/ID approach. Intuitively, we expect the gain transistors to be in the moderate inversion. The reason is that in weak inversion the dependency between the current and the V_{gs} of the device is exponential, and highly non-linear. On the other hand in strong inversion, the bias voltage is high meaning that the mobility degradation is causing non-linearities as well.

In order to achieve the compression without sacrificing a large amount of gain, it is important to bias the RF transistor where the IIP3 is maximum. The IIP3 is the crossing of the fundamental with the third harmonic, which creates a third order intermodulation product. This product exists in the frequency region of interest, which directly distorts our signal. By increasing the crossing point (IIP3) the third order intermodulation product decreases as it can be seen from equations 1 and 2.

$$IIP_3 = \sqrt{\frac{4}{3} \left| \frac{a_1}{a_3} \right|} = \sqrt{\frac{4}{3} \left| \frac{gm_1}{gm_3} \right|} \quad (1)$$

$$IM_3 = \frac{3}{4} \left| \frac{a_3}{a_1} \right| A^2 \quad (2)$$

Regarding my sizing strategy, I used a ration between the switching quad and the gm-transistors of 1/2. I tried to modify this ratio in the optimization process, but the improvement was not significant. The iteration process that I use to set the system before start optimizing is described below:

1. The conversion gain is an open parameter, thus I chose a large enough gain in order to minimize the non-linearities. The larger the gain, the smaller the input voltage swing, which result in smaller distortion considering that the IM3 product scales with V_{in}^3 .
2. After calculating the required gm to achieve the conversion gain goal, I sweep the width of the transistors respecting the ration I mentioned before. The goal is to get near the region of the required gm for my chosen conversion gain.
3. I plot the IIP3 plot and the transconductance plot in order to find the point which the IIP3 is maximized.
4. After setting the biasing, can find the common mode voltage of the switching transistors and their gate voltage swing by utilizing the equations from the theory slides. The LO switching quad was biased at 0.7V (random value) at the beginning.
5. After finishing the biasing and sizing of the transistors, I have to check if the saturation condition is violated. I perform a transient analysis to check if all the transistors are in saturation. Technics such as changing the bias point of the LO or slightly the width has to be performed in order to pull the transistors out of triode, if there are any. The new values of biasing or sizing have to be applied again in the DC simulation and the whole process has to be iterated.

In this section the schematic (given template), the g_m/ID plot and the theory equations are being presented. In figure 2, it can be clearly seen that the ideal biasing of the gm-transistors is at moderate inversion, as it was discussed before. The plot of the gm/ID has been performed after the full flow of the design.

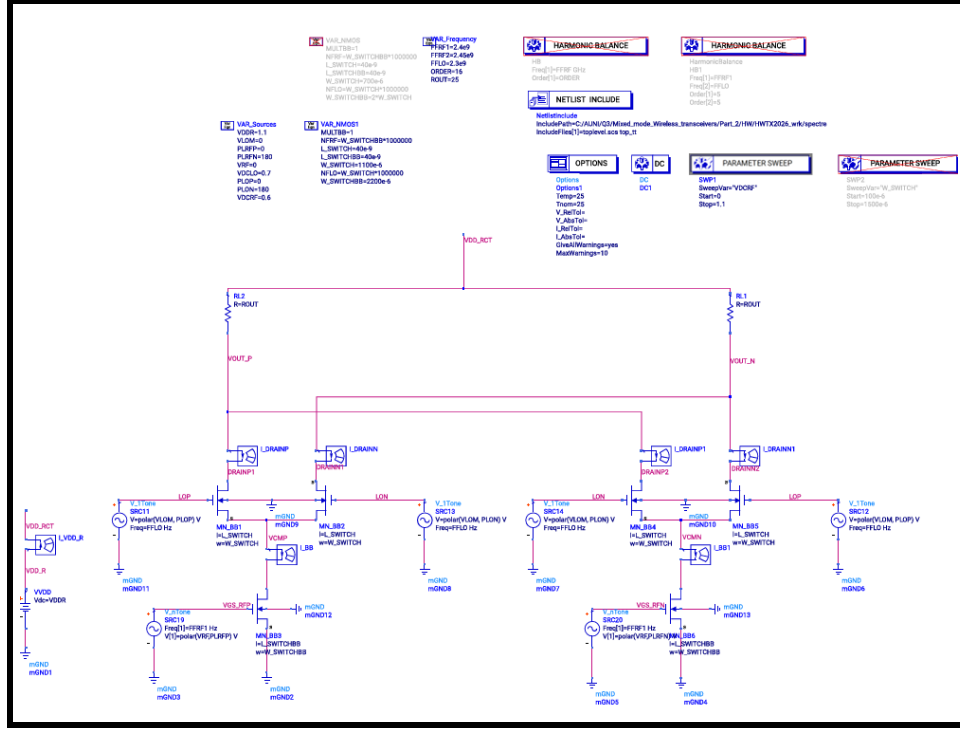


Figure 1: schematic.

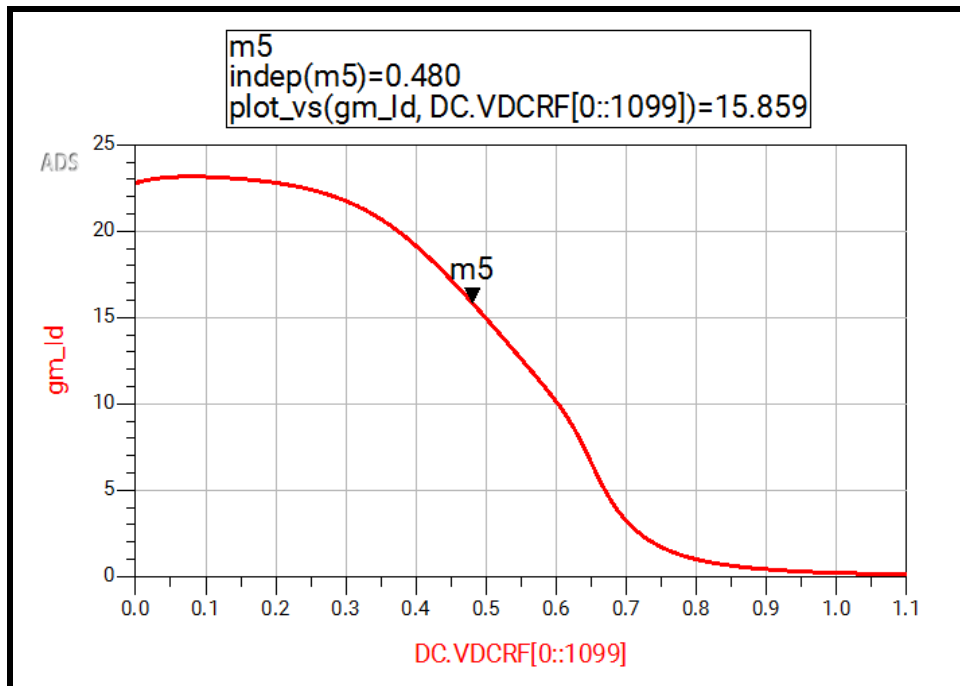


Figure 2: gm/ID.

The conversion gain of the double balanced active mixer is:

$$C_g = \frac{2}{\pi} gm R_D \quad (3)$$

Where $R_D = 25 \Omega$, and gm is the transconductance of the gain transistors. Moreover, the condition for the common mode voltage of the switching quad is:

$$V_{CM,LO} \geq V_{OV1} + V_{gs,LO}$$

Since we are aiming for the maximum possible gain, we want the drain node of the LO transistors to be as low as possible. This require the common mode voltage to be as low as possible, thus:

$$V_{CM,LO} = V_{OV1} + V_{gs,LO} \quad (4)$$

According to theory the LO amplitude which guarantees normal switching is:

$$V_O = \frac{\sqrt{2}}{2} (V_{gs,LO} - V_{tha}) \quad (5)$$

Furthermore, in order for the switching quad to remain in saturation the below condition should be met (it is important to remain in saturation or off; triode region causes more non-linearities).

$$V_{X,min} = V_{CM,LO} + \frac{\sqrt{2}}{2} (V_{GS,LO} - V_{TH,LO}) - V_{TH,LO} \quad (6)$$

Substituting the equation 4 to 6, the minimum drain voltage become:

$$V_{X,min} = V_{GS,RF} - V_{TH,RF} + \left(1 + \frac{\sqrt{2}}{2}\right) (V_{GS,LO} - V_{TH,LO}) \quad (7)$$

Moreover, the maximum voltage drop at the load in excursion is when both the current of the two gm-stages are applied to the load, thus:

$$V_{R,max} = 2 \cdot ID \cdot RD \quad (8)$$

Equation 7 can also express the maximum voltage drop if it is subtracted by the V_{DD} ; therefore, by combining the equation 7 and 8 we got:

$$2 \cdot ID \cdot RD = V_{DD} - \left| V_{GS,RF} - V_{TH,RF} + \left(1 + \frac{\sqrt{2}}{2}\right) (V_{GS,LO} - V_{TH,LO}) \right| \quad (9)$$

From equation 4, 5 and 9 the common mode and the voltage swing of the LO switching quad can be found.

2 Initial Designing phase [Transconductance and IIP3 plot/Questions 2 and 3]

Firstly, I want to stress that since there is no particular targeted conversion gain, I aim for the biggest possible conversion gain throughout the process, since it will reduce the required input swing and as a result the non-linearities. However, the selection of the biasing point is not entirely decided by this specification but actually dominated by the IIP3 curve. Therefore, during my design process the gain might deviate from the targeted value that I choose by some dB, but this is acceptable as soon as it is quite large and gives the maximum possible IIP3.

I choose a targeted conversion gain of 15dB. According to equation 3 the required transconductance is 0.35S. Firstly I set the ration of the LO and the RF transistors as 1/2, and I sweep the width to get close to this value **3**. The biasing of RF and LO transistors are 0.6V and 0.7V, respectively (random values).

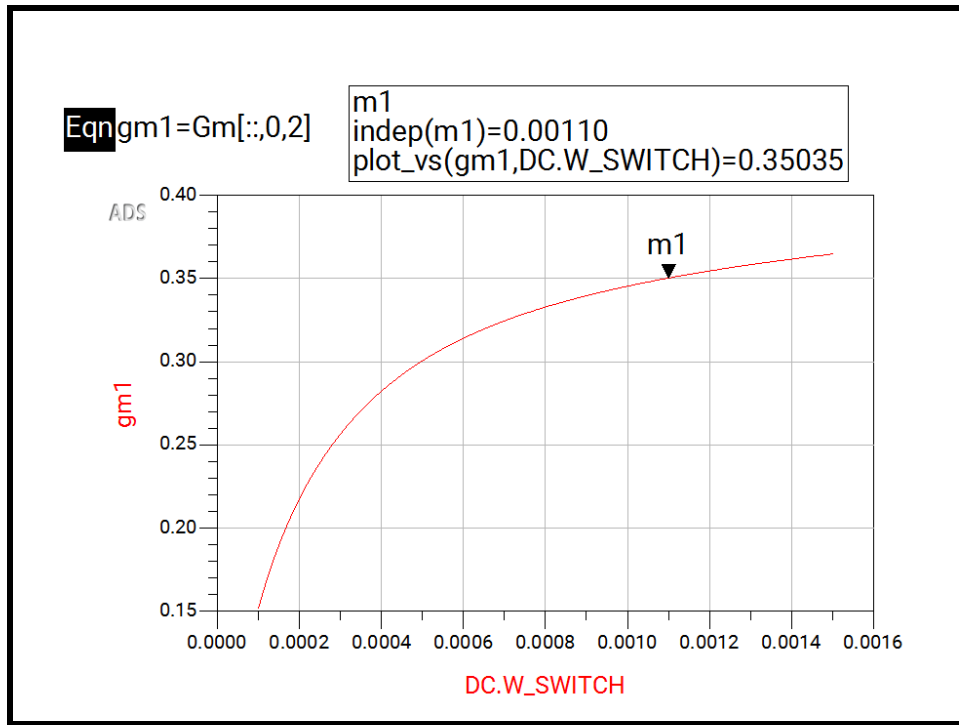


Figure 3: Sweeping the width of the transistors.

For $W_{RF} = 2200\mu m$ and $W_{LO} = 1100\mu m$, I achieve the transconductance value. Afterwards, I plot the transconductance plot and IIP3 (**4**, **5**) to find the biasing point of the gm-transistors which corresponds to maximum linearity.

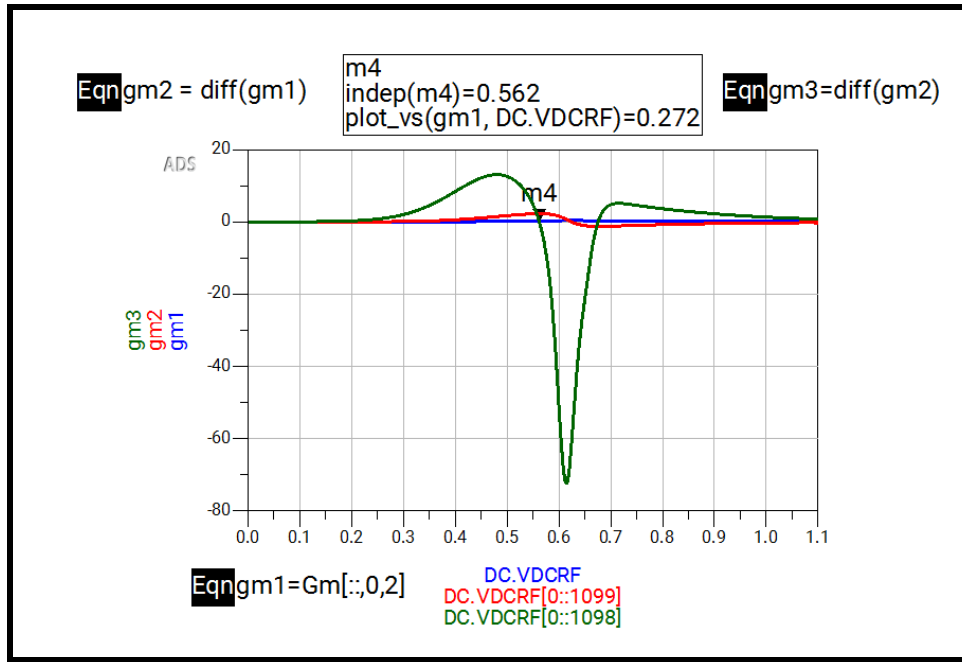


Figure 4: Transconductance plot.

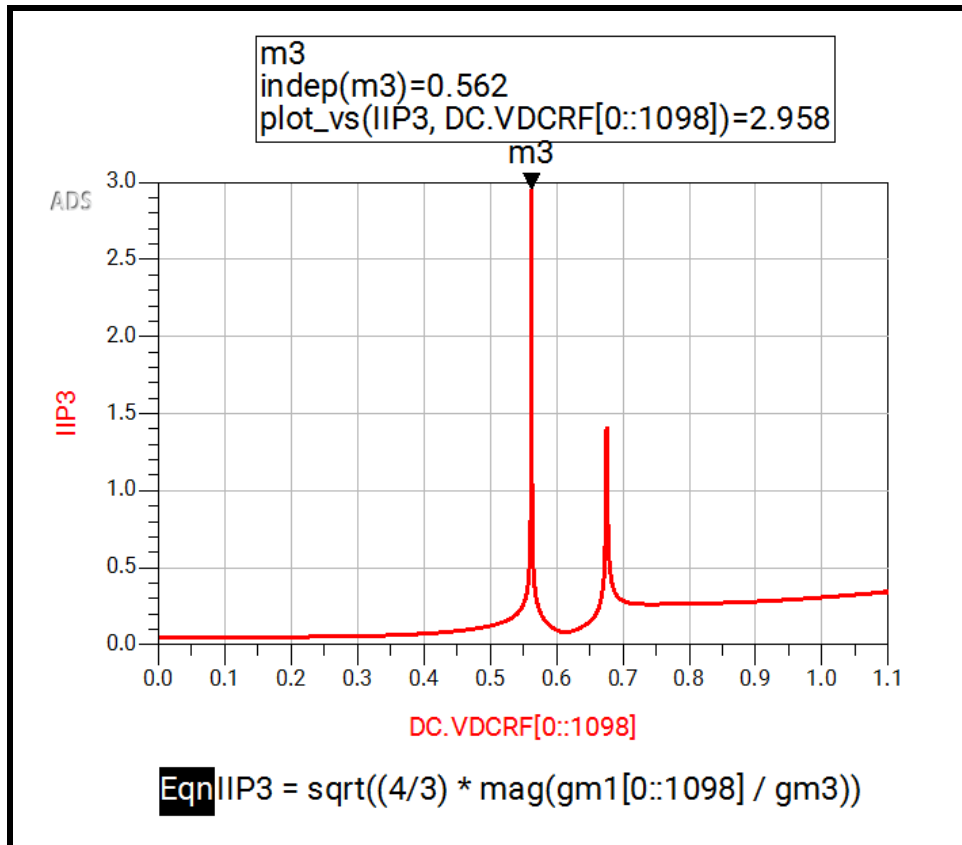


Figure 5: IIP3.

At this moment the biasing point of the RF transistor has changed to 0.562V which gives a transconductance of 0.272S. Now that I got the biasing point, I can calculate the common mode of the switching quad and the amplitude of the LO signal through

equations 4, 5 and 9. However, the threshold voltages of the transistors have to be extracted first **6**.

index	Vth
1	0.641
2	0.641
3	0.636
4	0.641
5	0.641
6	0.636

Figure 6: Vth.

3 Transient analysis [Question 4]

The resulted common mode and the LO amplitude from the previous section are:

$$V_{CM} = 698mV \quad V_O = 92.78mV$$

I perform a transient simulation with those values to inspect which of the transistors are out of saturation. The required input power for this simulation is computed from the conversion gain. Specifically from the output power specification:

$$4dBm = 10\log\left(\frac{V^2_{peak, out}}{2 * 50 * 1mW}\right) \Rightarrow$$

$$V_{peak, out} = 0.5V$$

Since the transconductance that I got in the previous section was 0.272S, the conversion gain is approximately 12.73 dB. Therefore:

$$12.73dB = 20\log\left(\frac{V_{out, peak}}{V_{in, peak}}\right) \Rightarrow V_{in, peak} = 115.5mV$$

Which for a differential input is:

$$V_{in, peak} = 55mV$$

I check the saturation condition of all the transistors. In figure **7** it is clear that all the transistors are in saturation.

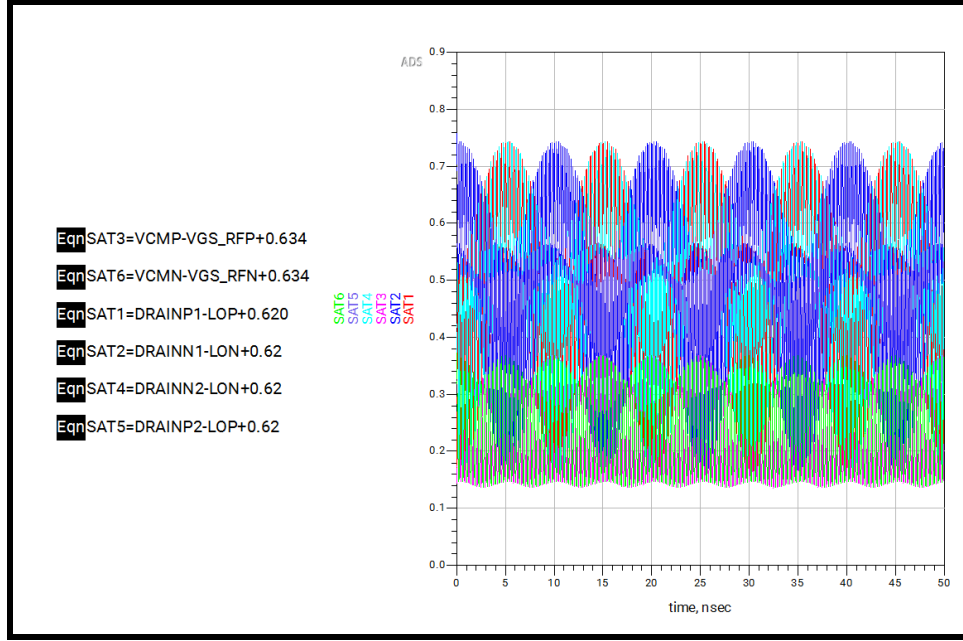


Figure 7: Saturation conditions.

4 Meeting the Specifications/Optimization [Question 7]

The questions 5 and 6 will be presented afterwards, since I wanted to plot the resulted gain after meeting the specifications. The starting point of the optimization is presented in figure 8. I applied the following techniques in order to get close to the specification.

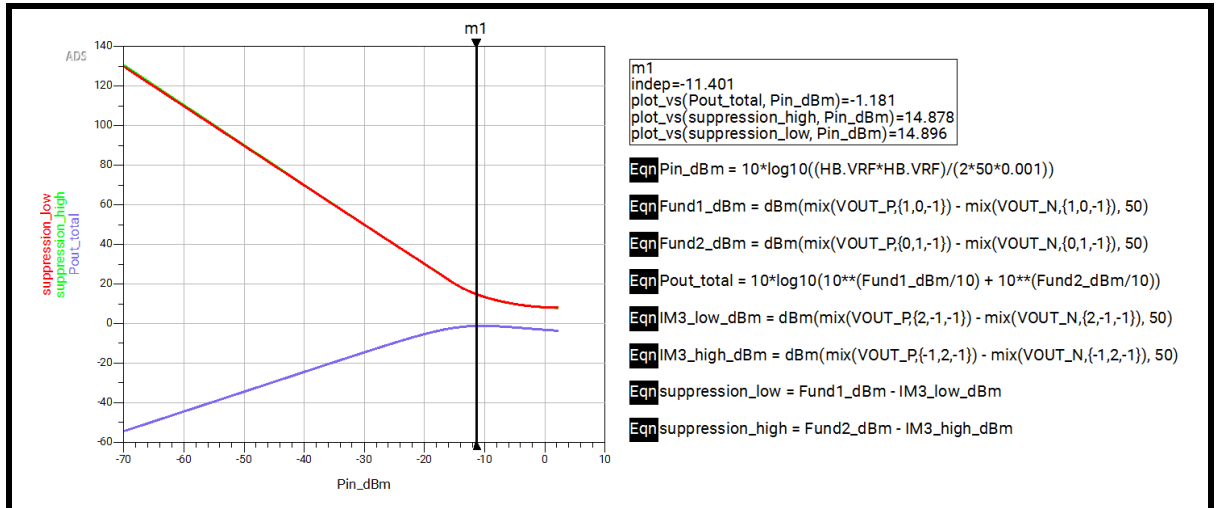


Figure 8: Output power and compression before optimization.

1. Firstly, I noticed that the V_O amplitude of the LO transistors was small. This is not ideal, considering that the switching of the switching quad has to be abrupt. The more abrupt the switching the greater the gain 10. However, I noticed that increasing the LO amplitude drags the switching quad out of

saturation, thus I reduced also the common mode voltage to compensate that change.

$$A_v = \frac{2}{\pi} g_{m1} R_D \left[1 - \frac{(V_{GS} - V_{TH})}{5\pi V_{p,LO}} \right] \quad (10)$$

2. I noticed that while the suppression drops for higher input voltage swing, which is expected considering that the non-linearities increase with the rise of the input swing, there is a slight peak in this behaviour. Therefore, by changing the biasing of the RF transistor (specifically decreasing it) you can move this optimum point to higher input voltage swings, achieving both better gain and linearity.

The resulted output power and compression plot is presented in figure 9. As it is visible, the compression has been achieved, but the output power is 1.5 dB less approximately.

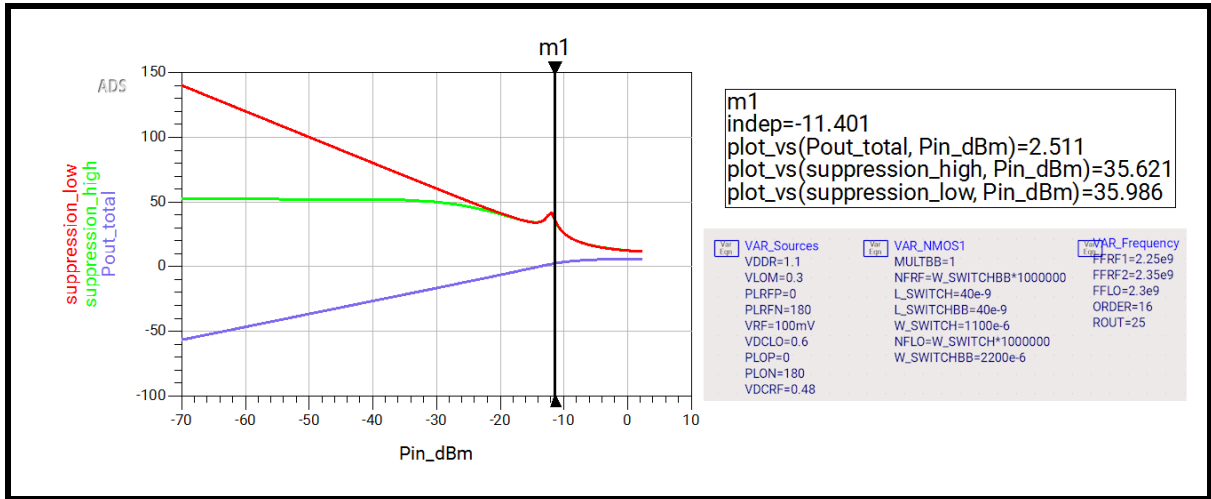


Figure 9: Output power and compression after optimization.

Lastly, I check if all the transistors are in saturation. In figure 10, those conditions are met.

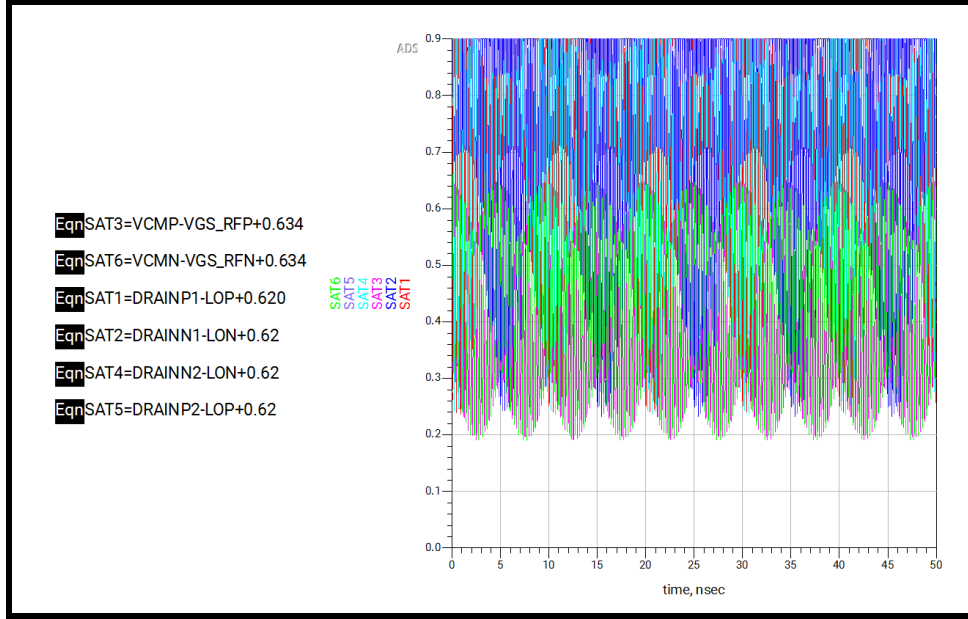


Figure 10: Saturation Conditions.

5 Plotting conversion gain over the LO amplitude and the input power [Questions 5 and 6]

Having optimized the design, I am proceeding to plot the conversion gain over the LO amplitude and the input power respectively. The conversion gain over LO amplitude is presented below:

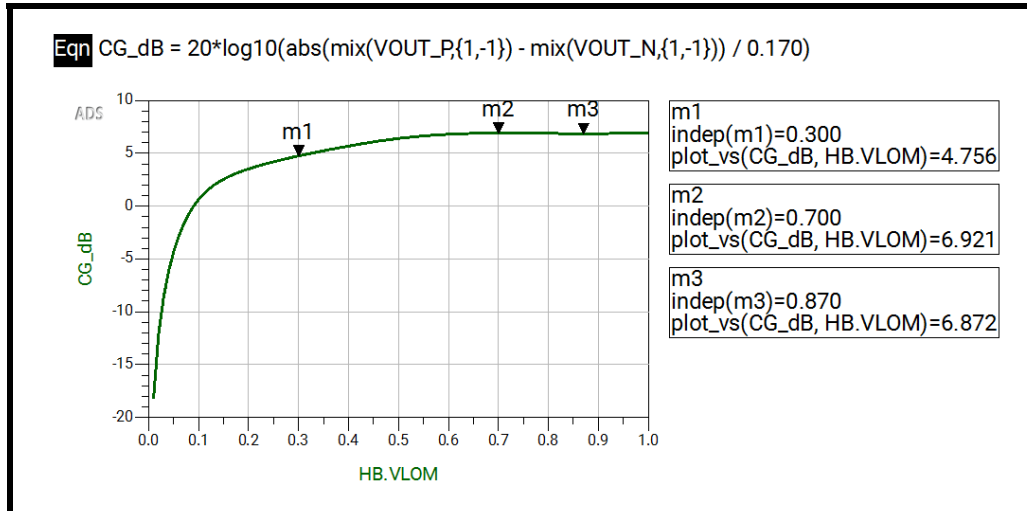


Figure 11: Conversion gain over LO amplitude.

As described in the previous section, the LO switching quad should switch abruptly enough in order to not degrade the conversion gain. However, really large LO swing can drug the transistors out of saturation, contributing to non-linearities, while achieving not significant amounts of conversion gain. This behaviour can also be seen from figure 11. The transition boundary between the soft and hard switching

can be seen just before the 0.2V. After the 0.2V the slope is decreased and the gain starts to saturate. Actually for really large voltage swings it is also decreasing as it is noted by the markers m2 and m3. Lastly, the conversion gain is significantly smaller after the optimization but the results are close to the specifications.

$$C_G = 4.756\text{dB}$$

The plot of the conversion gain over the input power is presented in figure 12. The marked point is referring to the input voltage swing that achieves the abovementioned specifications. As it becomes apparent the resulted gain is the same as the previous plot.

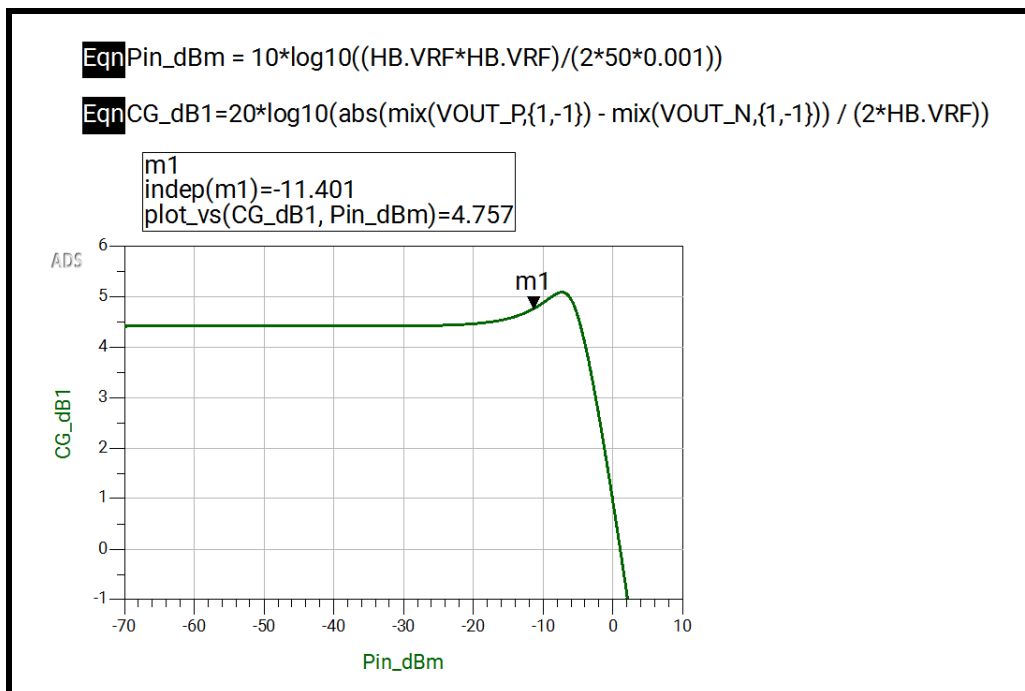


Figure 12: Conversion gain over LO amplitude.

6 Output power versus the input power and conversion point

In figure 13 I plot the output power versus the input power and I mark the 1dB conversion point. It is worth mentioning that the +6 term in the expression of the ideal output power comes from the fact that the input power that I calculated is single ended while the output power is differential. Lastly, the +4.411 term is the conversion gain at low input power, which as we saw in the previous section 12 remained constant.

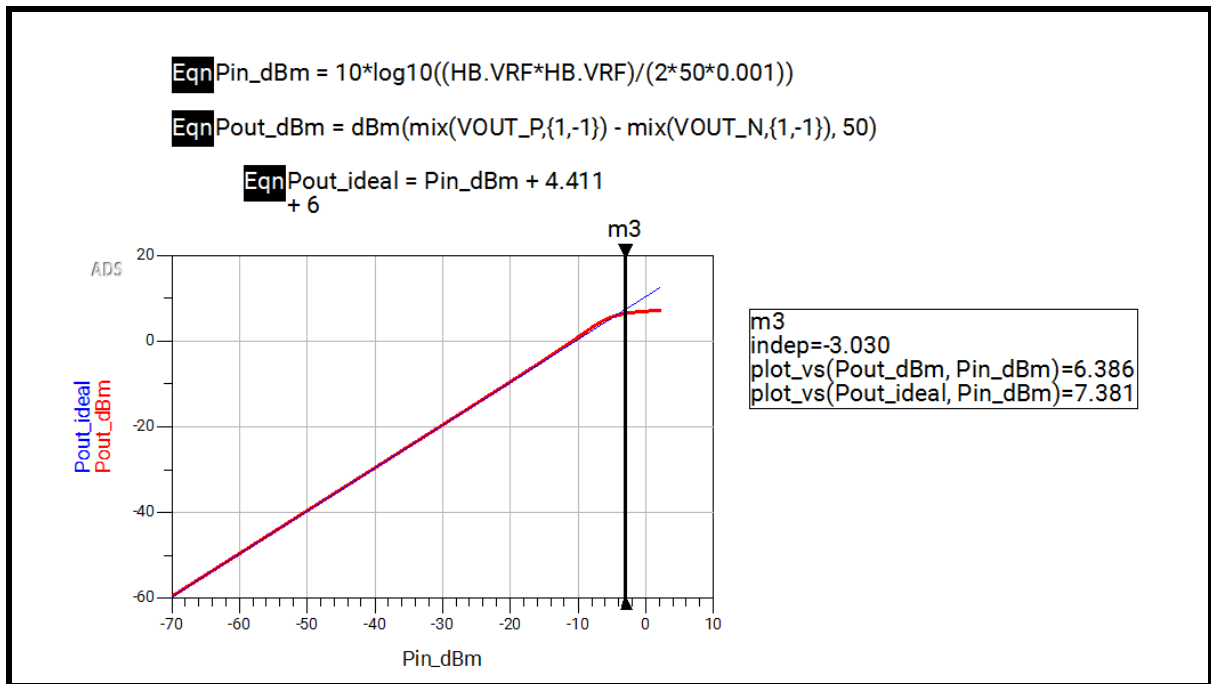


Figure 13: 1dB compression point.